INDEX

Daffodil Institute of IT

Lab Report
On
DSD-510209
(Digital System Design Lab)

Experiment No.	07
Experiment Name	Verify the operation of basic gates
Date of Performance	
Date of Submission	

Submitted By

Name	**Abdur Rahim**
Student ID	**180121**
Registration No	
Section	**B**
Batch	**18th**

Submitted To

Tahmina Aktar Trisha
Faculty
Department of Computer Science & Engineering
Daffodil Institute of IT(DIIT)
64/3 & 64/4 Lake Circus (2nd-5th floor)
Mirpur Road Dhaka-1205, Bangladesh

Signature

Expriment NO : 01

Expriment NAME: Verify the operation of basic gates.

Objective: The objective of the expriment is to verify the operation of the basic gate and also verify the truth table of basic gate using speific IC

Operation of basic gates

Basic gate: In electronics, a logic is an idealized or physical device implementing a boolean function that is it performs a logical operation on one or more binary inputs and produces a single binary output. Threes basic gates are,

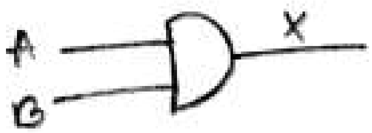
i) AND
ii) OR
iii) NOT

AND gate: The AND operation describe as when two input variasle are in high position or in logic 1, then the resulted autput is 1. If input is low or logic 0 the autput will se 0.

Truth table:

A	B	A.B = X
0	0	0
0	1	0
1	0	0
1	1	1

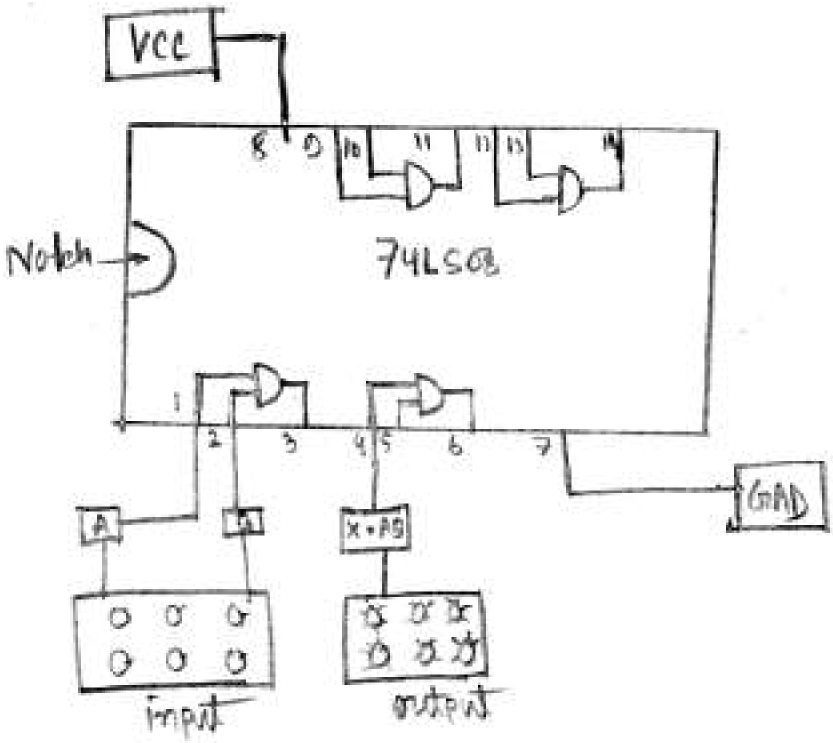
Logical symbol :



Boolean expression of AND gate:

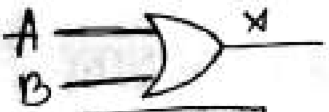
Instrument:
i) voltage source (+3V - +12V)
ii) trainer Board
iii) Ic (74LS08)
iv) Bread Board
v) cable
vi) switch
vii) LeD
viii) connecting wire

④ Circuit specifications



AND gate

④ OR gate:



$\boxed{X = A+B}$

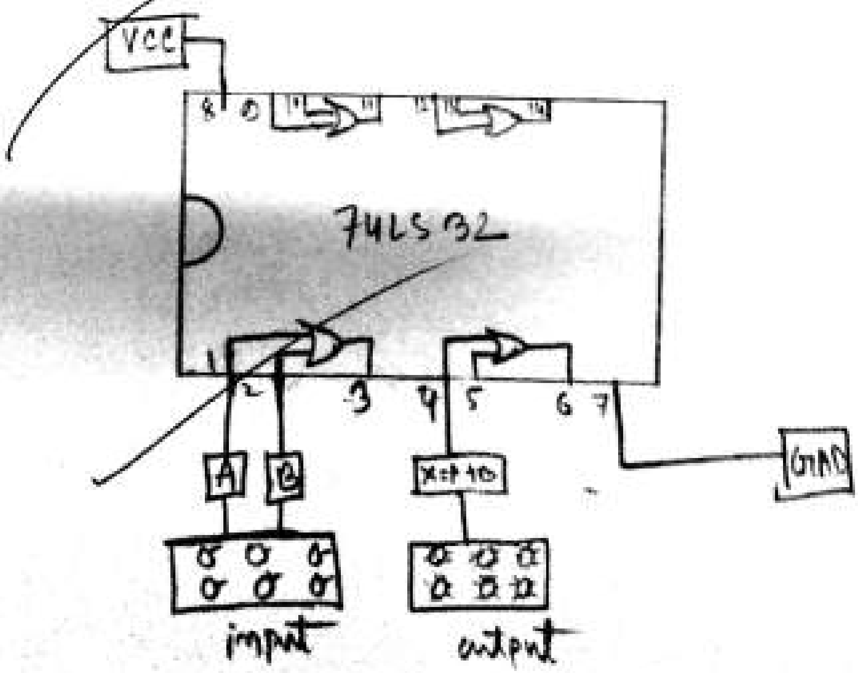
Boolean expression.

A	B	$x = A+B$
0	0	0
0	1	1
1	0	1
1	1	1

OR expression OR OR operation describe as when two input variable are low position the resulted output is high resulted output is 1

Instrument :-
i) Voltage source
ii) led
iii) switch
iv) cable
v) Bread board
vi) Ic (74LS32)
vii) Trainer board

circuit specification:



OR GATE

NOT gate: NOT gate is just inverse operation. NOT operation perform into simple variable. If a variable is declare as x of A then the result of x of A can be expresses as $x = \overline{A}$.

Input	output
0	1
1	0

Truth table

Boolean expression of NOT gate:

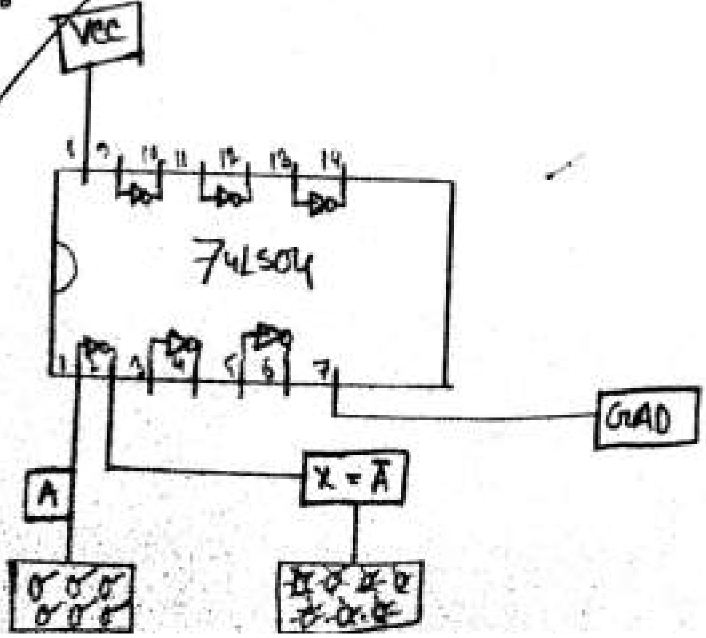
A	$x = \overline{A}$
0	1
1	0

$$X = \text{NOT } A$$
$$= \overline{A}$$

circuit symbol of NOT gate

Equipment:
i) voltage source
ii) Trainer board
iii) IC (74LS08, AND), (74LS04, NOT), (74LS32, OR)
iv) Bread board
v) cable
vi) switch
vii) LED

Circuit specification:-



Result : By this experimental, we can know to implement Basic gate

NOT →

A	$\bar{A}$	
0	1	→ LED ON
1	0	→ LED off

OR →

A	B	A+B	
0	0	0	→ LED off
0	1	1	→ LED on
1	0	1	→ LED on

AND →

A	B	AB	
0	0	0	→ led off
0	1	0	→ led off
1	0	0	→ led off
1	1	1	→ led on

Discussion : After the expriment we know how logic Basic gates work on the basic Boolean-Algebra. We can construct any other logic gate using its combination in a perticular way.

Daffodil Institute of IT

Lab Report
On
DSD-510209
(Digital System Design Lab)

Experiment No.	02
Experiment Name	Verify the universality of NAND and NOR gate
Date of Performance	
Date of Submission	

Submitted By

Name	**Abdur Rahim**
Student ID	**180121**
Registration No	
Section	**B**
Batch	**18th**

Submitted To

Tahmina Aktar Trisha
Faculty
Department of Computer Science & Engineering
Daffodil Institute of IT(DIIT)
64/3 & 64/4 Lake Circus (2nd-5th floor)
Mirpur Road Dhaka-1205, Bangladesh

Signature

Experiment number : 02

Experiment name :- Varify the universality of NAND and NOR gate

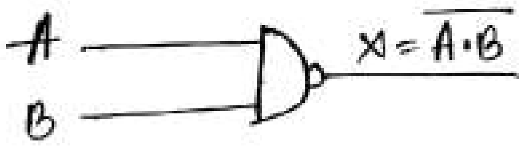
Objective : The objective of this experiment is to verify the universality of NAND and NOR gate to implement the operation of basic gate.

Operation of universal gate : A universal gate is a gate which can implement any Boolean function without using any other gate type. The NAND and NOR gates are universal gate.

NAND gate:- It is the similar like AND gate symbol except for the small circle on its output. This small circle denote as inversion operation. Thus the NAND operate like AND gate.

Boleam expression of NAND gate is

$$X = \overline{A \cdot B}$$



logical symbol

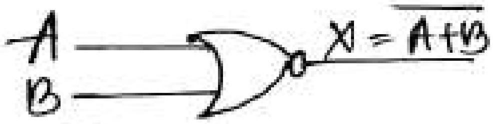
A	B	A·B	$X = \overline{A \cdot B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

truth table

4) NOR gate: It is similar like OR gate symbol except for the small circle on its outputs. The small circle denotes as inversion operation. Thus the NOR operates like OR gate.

Boolean expression of NOR gate is,

$$X = \overline{A+B}$$



Logical symbol

A	B	A+B	$X=\overline{A+B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

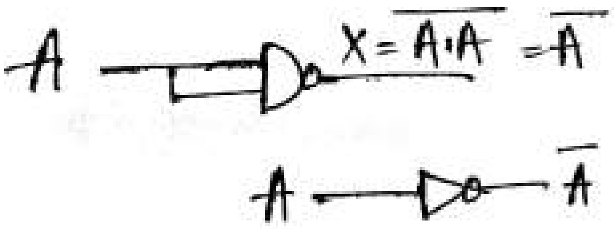
Truth table

Equipment:

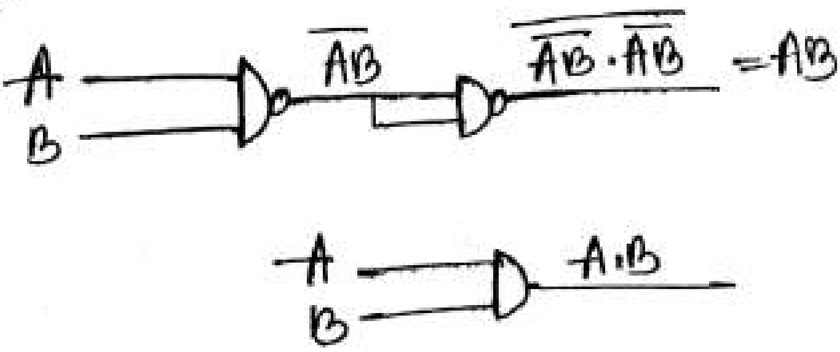
i) power supply
ii) Ic (74LS00, NAND), (74LS02, NOR)
iii) Bread board
iv) cable
v) switch
vi) LED
vii) connecting wire

Verify the universality of NAND using logical circuit:

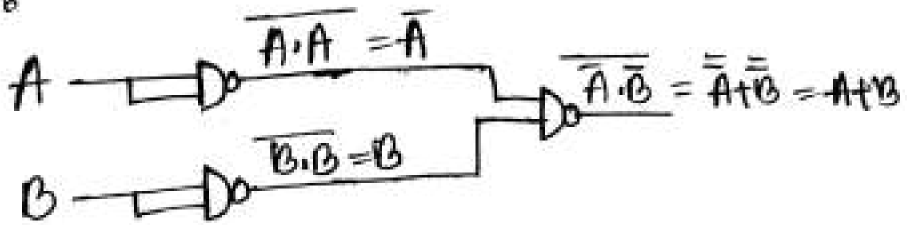
NAND to NOT:

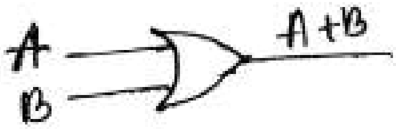


NAND to AND:



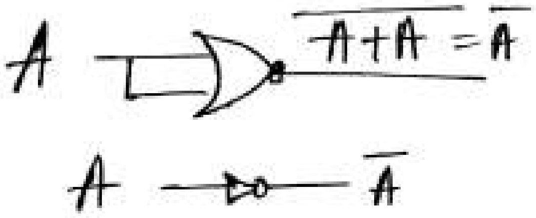
NAND to OR :



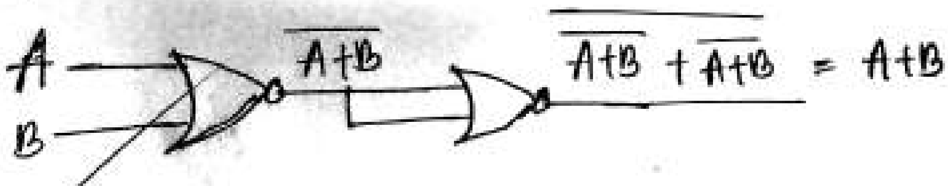


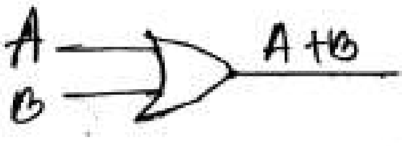
Verify the universality of NOR gate using logical circuit :

NOR to NOT :



NOR to OR :





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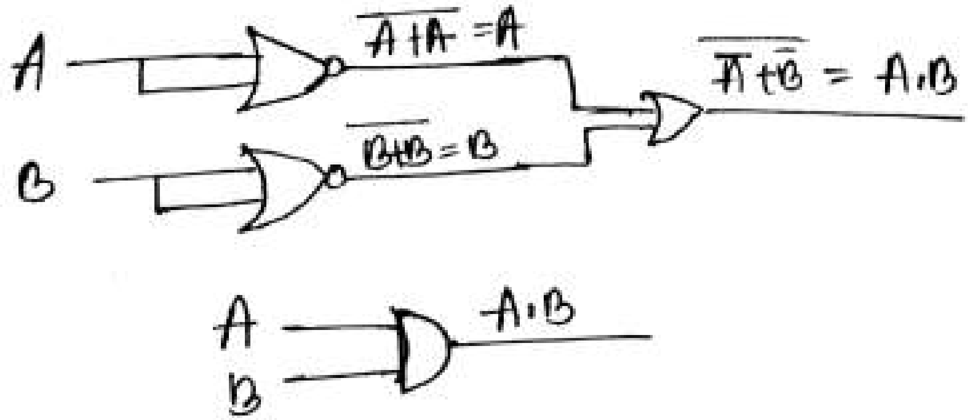
④ NOR to AND :



④ circuit specification :

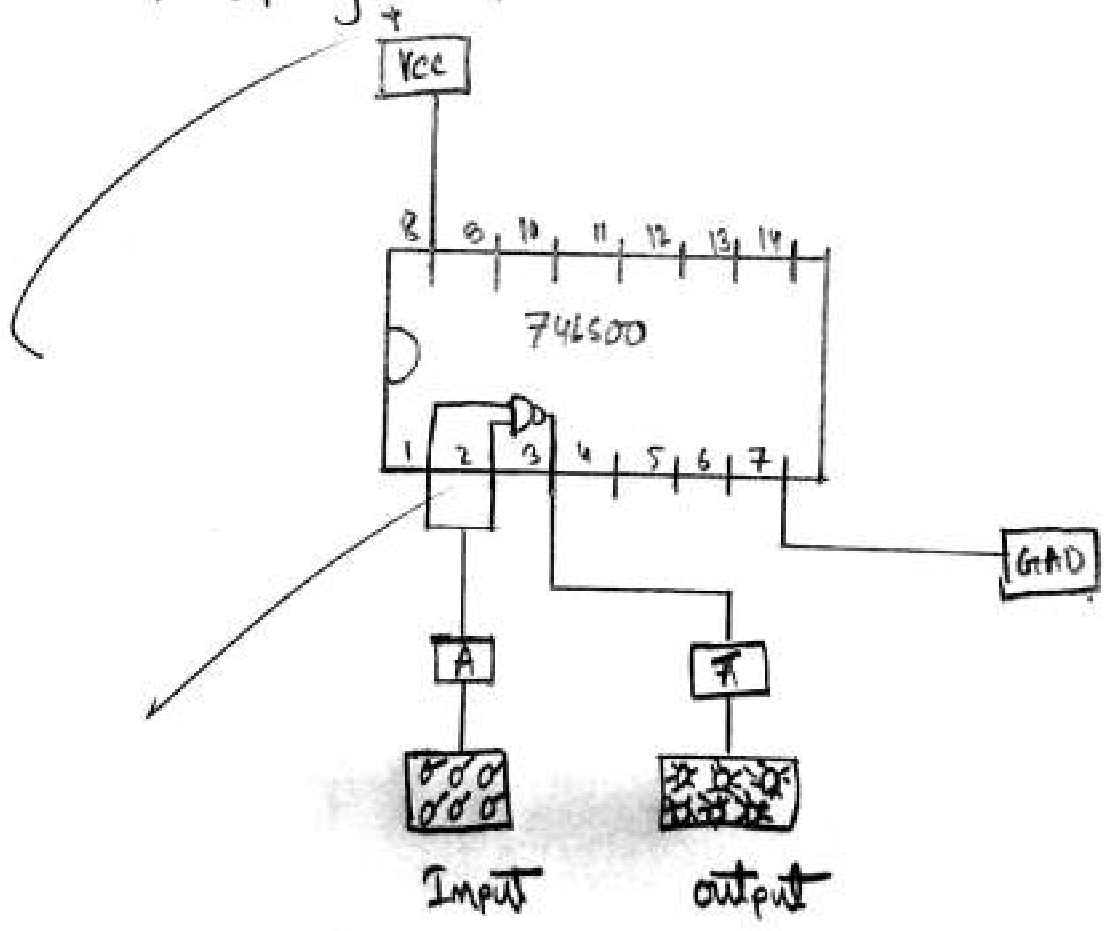


Fig : FOR NAND to NOT

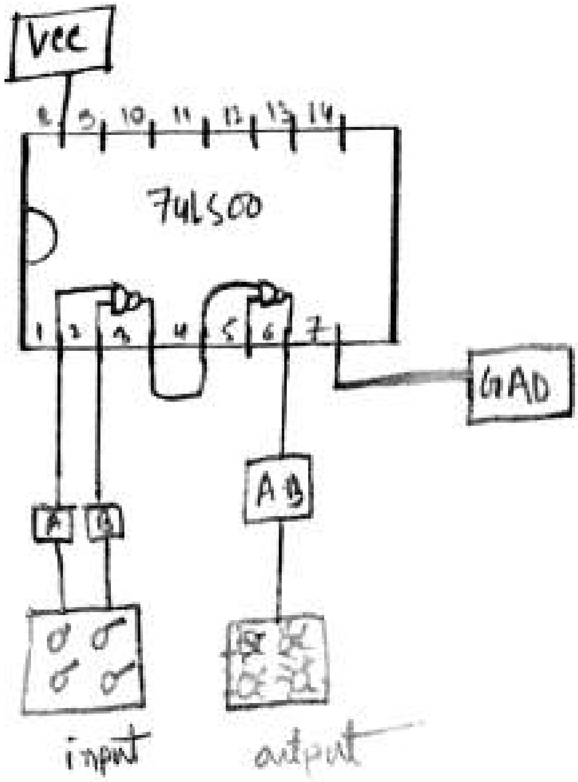


Fig: circuit for NAND to AND

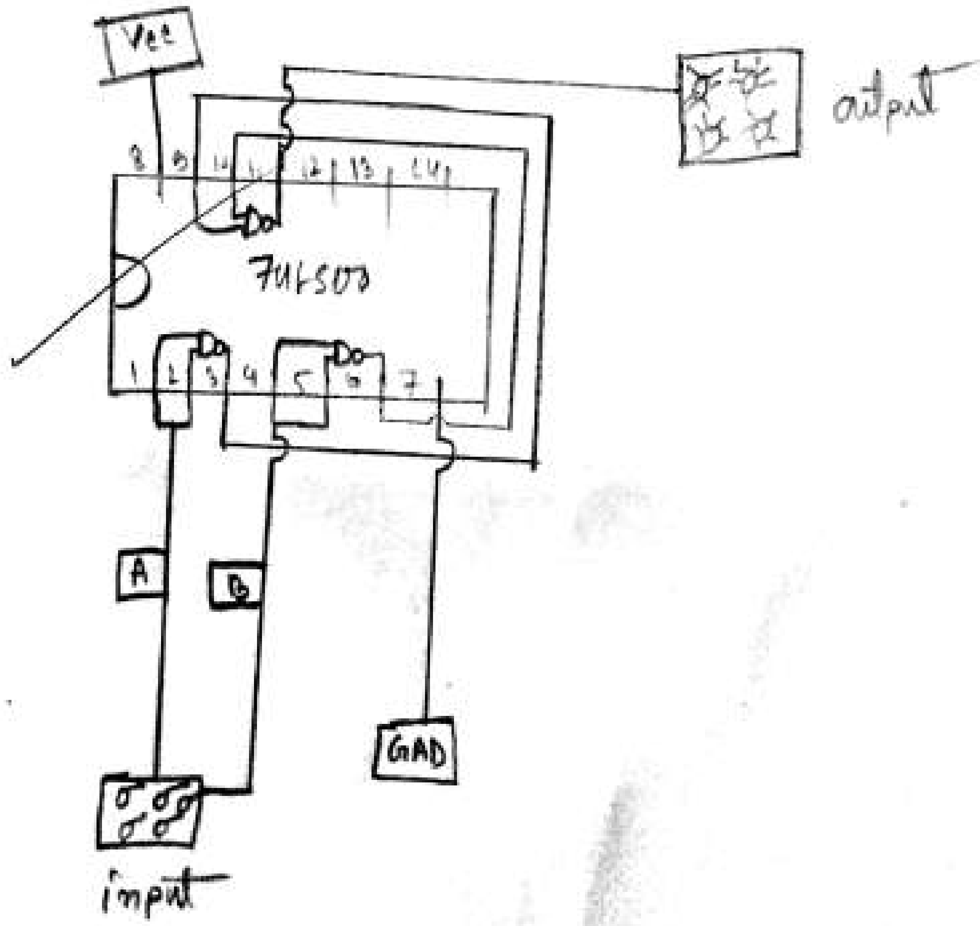


Fig: NAND to OR gate

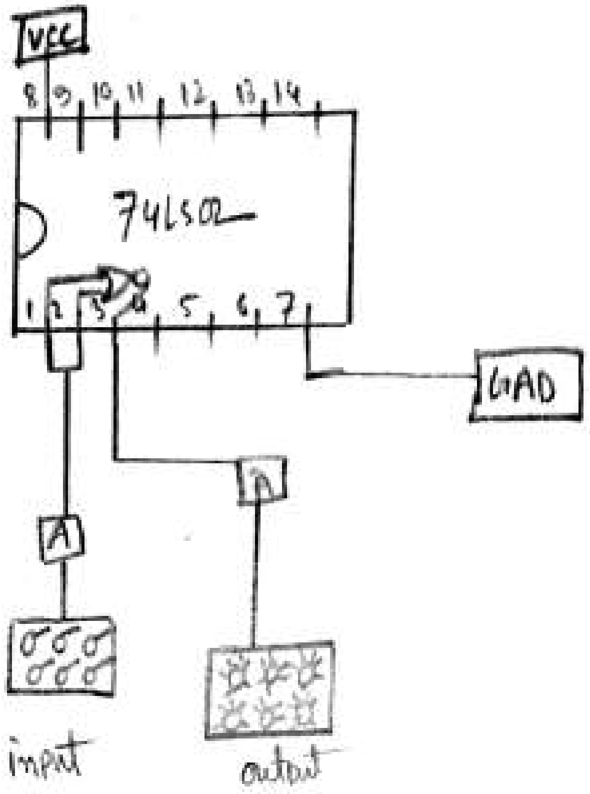


Fig : NOR to NOT gate

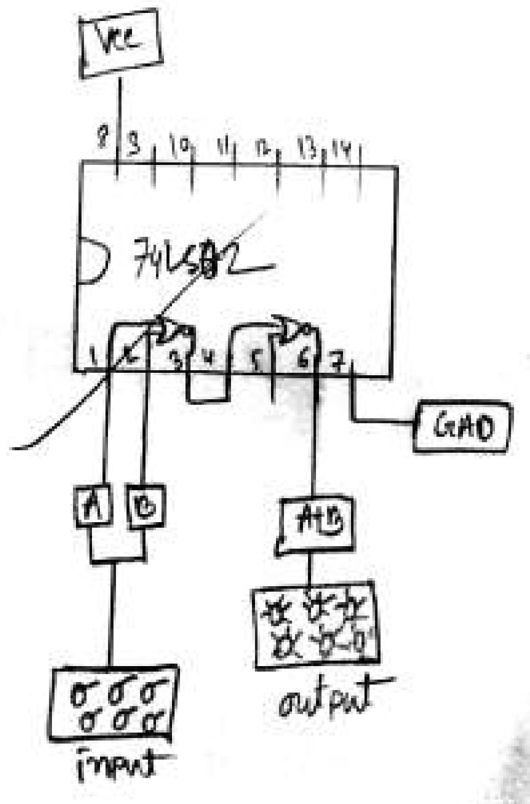


Fig : NOR to OR gate

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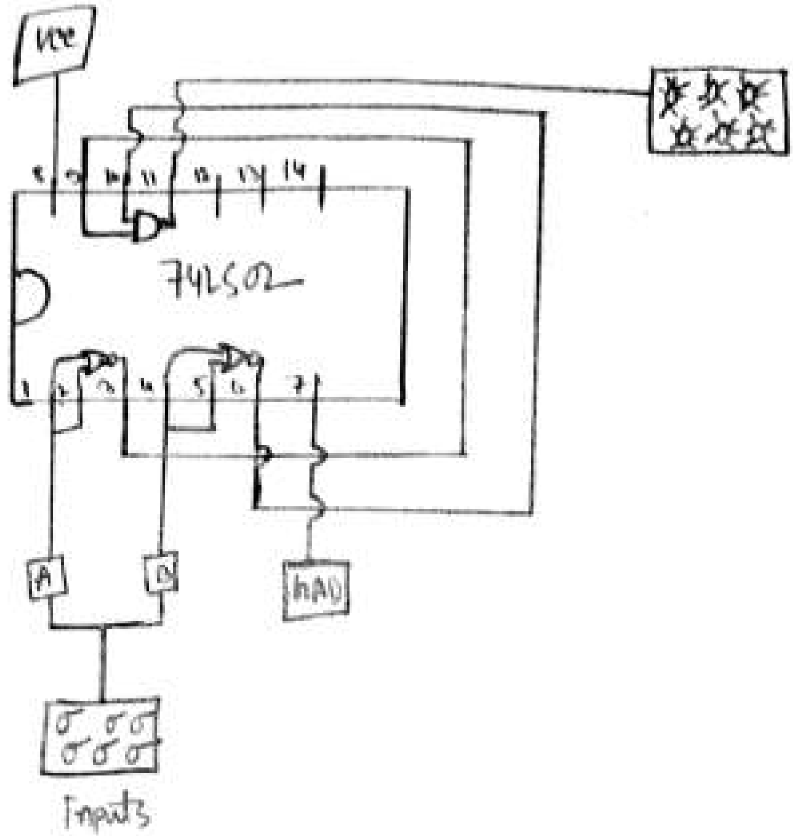

Fig: NOR to AND gate

Result for (NAND to NOT gate)

A	$\bar{A}$	LED
0	1	ON
1	0	OFF

Result for NAND to AND gate:

A	B	A.B	LED
0	0	0	OFF
0	1	0	OFF
1	0	0	OFF
1	1	1	ON

Result for NAND to OR gate:

A	B	A+B	LED
0	0	0	OFF
0	1	1	ON
1	0	1	ON
1	1	1	ON

Result for NOR to NOT gate:

A	$\overline{A}$	LED
0	1	ON
1	0	OFF

Result for NOR to OR gate

A	B	X=A+B	LED
0	0	0	OFF
0	1	1	ON
1	0	1	ON
1	1	1	ON

Result for NOR to AND gate :

A	B	X = AB	LED
0	0	0	OFF
0	1	0	OFF
1	0	0	OFF
1	1	1	ON

Discussion

From this experiment we can verify the universality of NAND and NOR gates and also verify the truth table of this gate using different ICs

Daffodil Institute of IT

Lab Report
On
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Experiment No.	03
Experiment Name	Verification of operation $X = \bar{A}B + A\bar{B}$
Date of Performance	
Date of Submission	

Submitted By

Name	**Abdur Rahim**
Student ID	**180121**
Registration No	
Section	**B**
Batch	**18th**

Submitted To

Tahmina Aktar Trisha
Faculty
Department of Computer Science & Engineering
Daffodil Institute of IT(DIIT)
64/3 & 64/4 Lake Circus (2nd-5th floor)
Mirpur Road Dhaka-1205, Bangladesh

Signature

Expriment No: 3

Expriment Name: Verification the operation $X = \bar{A}B + A\bar{B}$

Objective: The objective of this experiment is to verify the operation of $X = \bar{A}B + A\bar{B}$ using three differents ICs

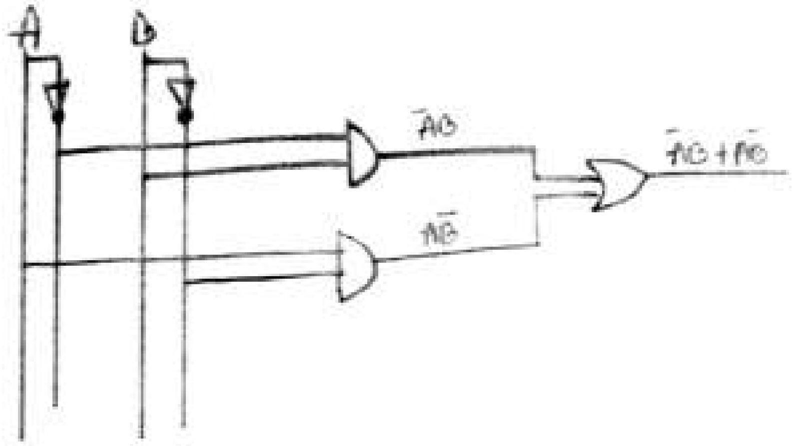
Operation of Ex-OR gate:

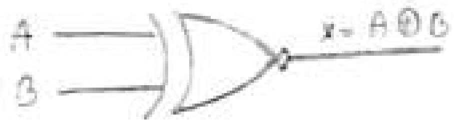
An Ex-OR gate is a digital logic gate with two or more inputs and one output. That performs exclusive disjunction. The output of XOR gate is true only when exactly one of its inputs is true.

Boolean expression of Ex-OR gate:

$$X = \bar{A}B + A\bar{B} = A \oplus B$$

Circuit diagram :-





Truth table of XOR Gate :

A	B	$\bar{A}$	$\bar{B}$	$\bar{A}B$	$A\bar{B}$	$\bar{A}B+A\bar{B}$
0	0	1	1	0	0	0
0	1	1	0	1	0	1
1	0	0	1	0	1	1
1	1	0	0	0	0	0

Truth table of $x = \bar{A}B + A\bar{B}$

$x = A \oplus B$

Equipment :

i) IC (74LS04) NOT, (74LS08) AND, (74LS32) OR
ii) Switch
iii) LED
iv) wire
(v) Bread Board
(vi) Multimeter

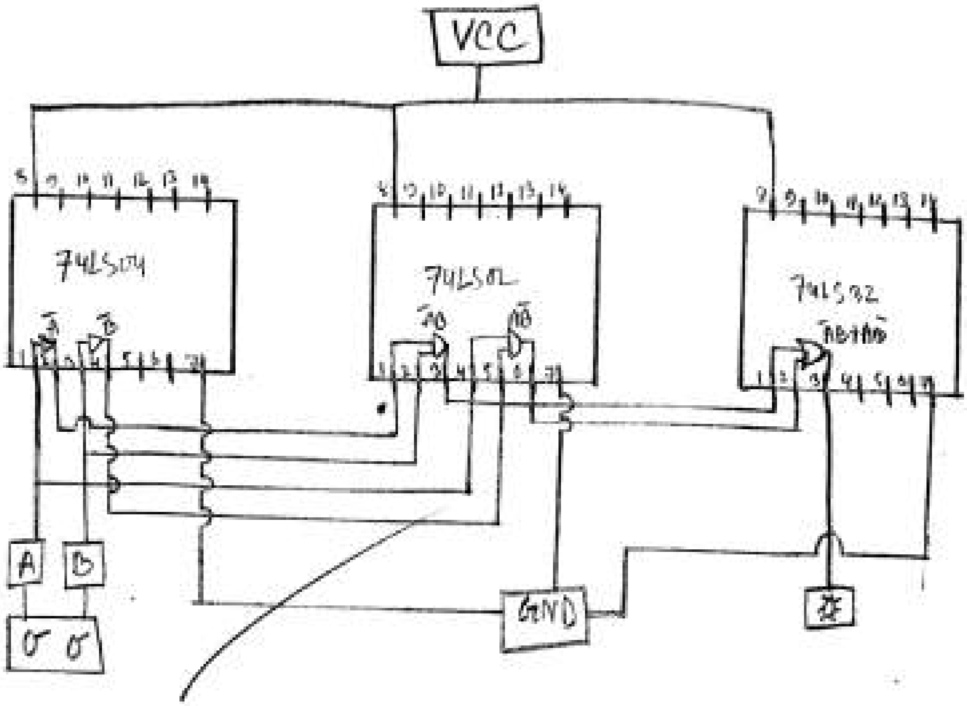


Fig : Circuit diagram

Results:-

A	B	$\bar{A}$	$\bar{B}$	$\bar{A}B$	$A\bar{B}$	$\bar{A}B+A\bar{B}$	LED
0	0	1	1	0	0	0	off
0	1	1	0	1	0	1	on
1	0	0	1	0	1	1	on
1	1	0	0	0	0	0	off

Discussion:— From this expriment we can verify the operation of Boolean expression $X = \bar{A}B + A\bar{B}$ using different IC's .

Daffodil Institute of IT

Lab Report On DSD-510209 (Digital System Design Lab)

Experiment No.	04
Experiment Name	Verify the operation of half adder circuit
Date of Performance	
Date of Submission	

Submitted By

Name	**Abdur Rahim**
Student ID	**180121**
Registration No	
Section	**B**
Batch	**18th**

LAB REPORT

Submitted To

Tahmina Aktar Trisha
Faculty
Department of Computer Science & Engineering
Daffodil Institute of IT(DIIT)
64/3 & 64/4 Lake Circus (2nd-5th floor)
Mirpur Road Dhaka-1205, Bangladesh

Signature

Experiment no: 04

Experiment name: Verify the operation of half adder circuit.

Objective: The objective of this experiment is to verify the half adder circuit to impliment of the AND and exclusive OR gate, using specific IC's

Operation of half adder circuit:-

A half adder circuit is a circuit perfroms the addition of numbers. The half adder is able to add two single binary digits and provide the outputs a carry value. It has two inputs (A,B) and outputs (S,C).

The common representation uses a XOR logic and an AND gate

Boolean expression of half adder

$$Sum : \bar{A}B + A\bar{B} = A \oplus B$$

$$C : AB$$

logical circuit :—

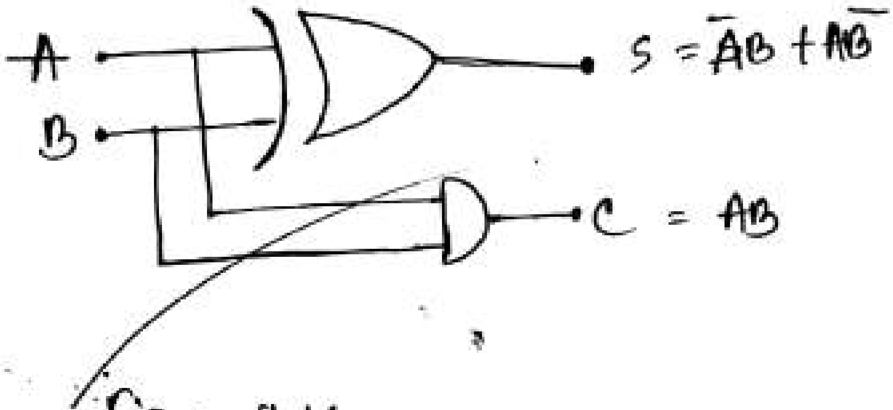


Fig : Half adder

Truth table :—

A	B	Sum	C AB
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

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The common representation uses a XOR logic and an AND gate

Boolean expression of half adder

$$\text{Sum} : \bar{A}B + A\bar{B} = A \oplus B$$

$$C : AB$$

logical circuit:-

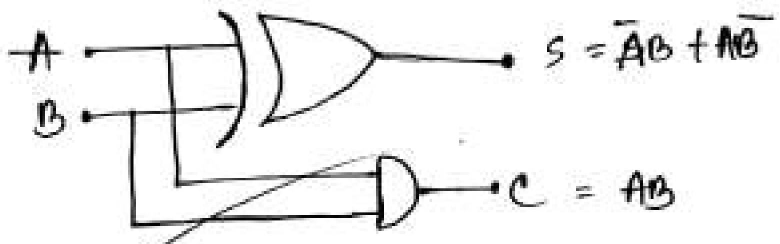

Fig: Half adder

Truth table:-

A	B	Sum	C AB
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Equipment :- i) Ic (74LS86) X-OR, (74LS08) AND

ii) Bread Board

iii) wire

iv) LED

v) Multimeter

vi) voltage source (+5v - +12v)

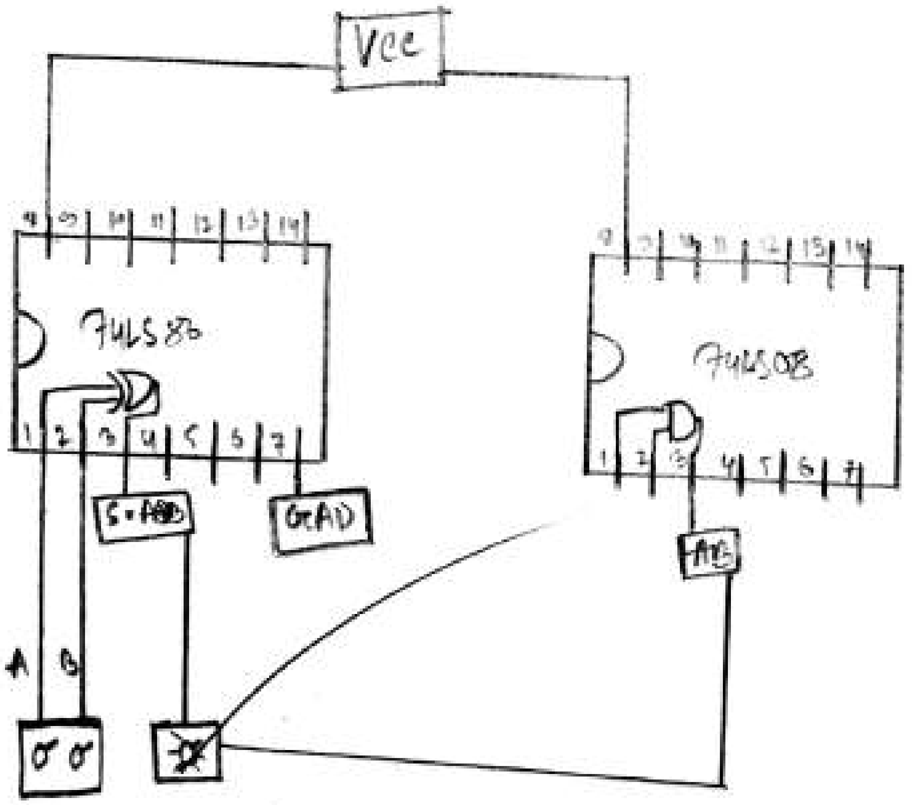

fig: Half adder

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Result:

A	B	$\bar{A}$	$\bar{B}$	$\bar{A}B$	$A\bar{B}$	$\bar{A}B+A\bar{B}$	LED	AB	LED
0	0	1	1	0	0	0	off	0	off
0	1	1	0	1	0	1	on	0	off
1	0	0	1	0	1	1	on	0	off
1	1	0	0	0	0	0	off	1	on

Discussion:-

From this experiment we can verify the operation of Boolean expression $\bar{A}B+A\bar{B}$ and carry AB using 74LS86 (X-OR) and 74LS08 (And) IC's

Daffodil Institute of IT

Lab Report
On
DSD-510209
(Digital System Design Lab)

Experiment No.	05
Experiment Name	Verify the operation of full adder circuit
Date of Performance	
Date of Submission	

Submitted By

Name	**Abdur Rahim**
Student ID	**180121**
Registration No	
Section	**B**
Batch	**18th**

Submitted To

Tahmina Aktar Trisha
Faculty
Department of Computer Science & Engineering
Daffodil Institute of IT(DIIT)
64/3 & 64/4 Lake Circus (2nd-5th floor)
Mirpur Road Dhaka-1205, Bangladesh

Signature

Experiment no: 05

Experiment name: Verify the operation of full adder circuit.

Objectives: The objectives of this experiment is to verify full adder circuit to implement of the AND gate, OR gate and EX-OR gate, using specific ICs

Operation of full adder circuit: A full adder circuit have three inputs bit and produce two outputs. There are input A,B,Cin and produce Sum and Cout. So, we have configuration of sum of three input bits and produce S and Cout output.

Boolean expression of full adder:

$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$$

$$= \bar{A}(\bar{B}C_{in} + B\bar{C}_{in}) + A(\bar{B}\bar{C}_{in} + BC_{in})$$

$$= \bar{A}(B \oplus C_{in}) + A(\overline{B \oplus C_{in}})$$

$$= A \oplus B \oplus C_{in}$$

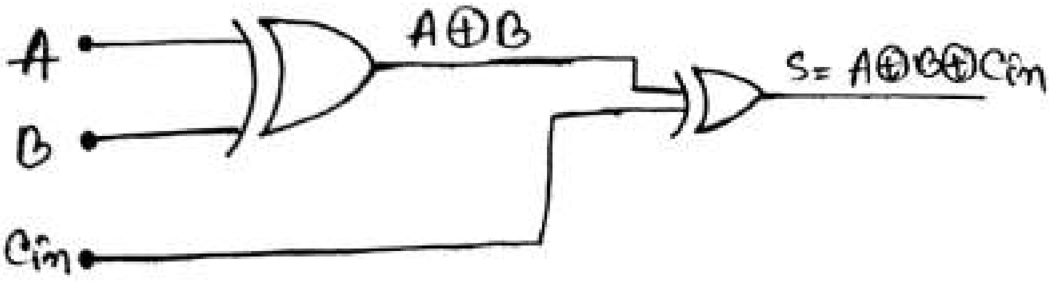
$$C_{out} = \bar{A}BC_{in} + A\bar{B}C_{in} + ABC_{in}$$

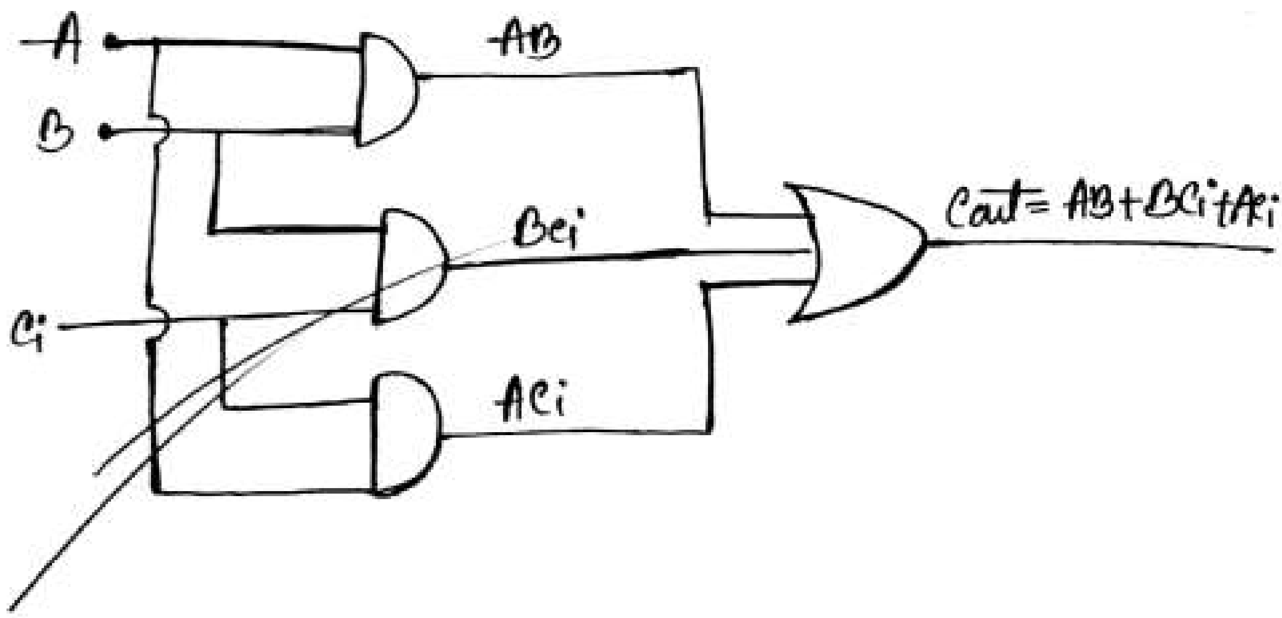
$$= \bar{A}BC_{in} + ABC_{in} + A\bar{B}C_{in} + ABC_{in} + AB\bar{C}_{in} + ABC_{in}$$

$$= BC_{in}(\bar{A}+A) + AC_{in}(\bar{B}+B) + AB(\bar{C}_{in}+C_{in})$$

$$= BC_{in} + AC_{in} + AB$$

Logical diagram :





Truth table:

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Apparatus:-
i) IC (74LS86) Ex OR, (74LS08) AND, (74LS32) OR
ii) Bread board
iii) Connecting wire
iv) switch
v) power supply
vi) cable
vii) LED
viii) Multimeter.

Circuit Diagram:-

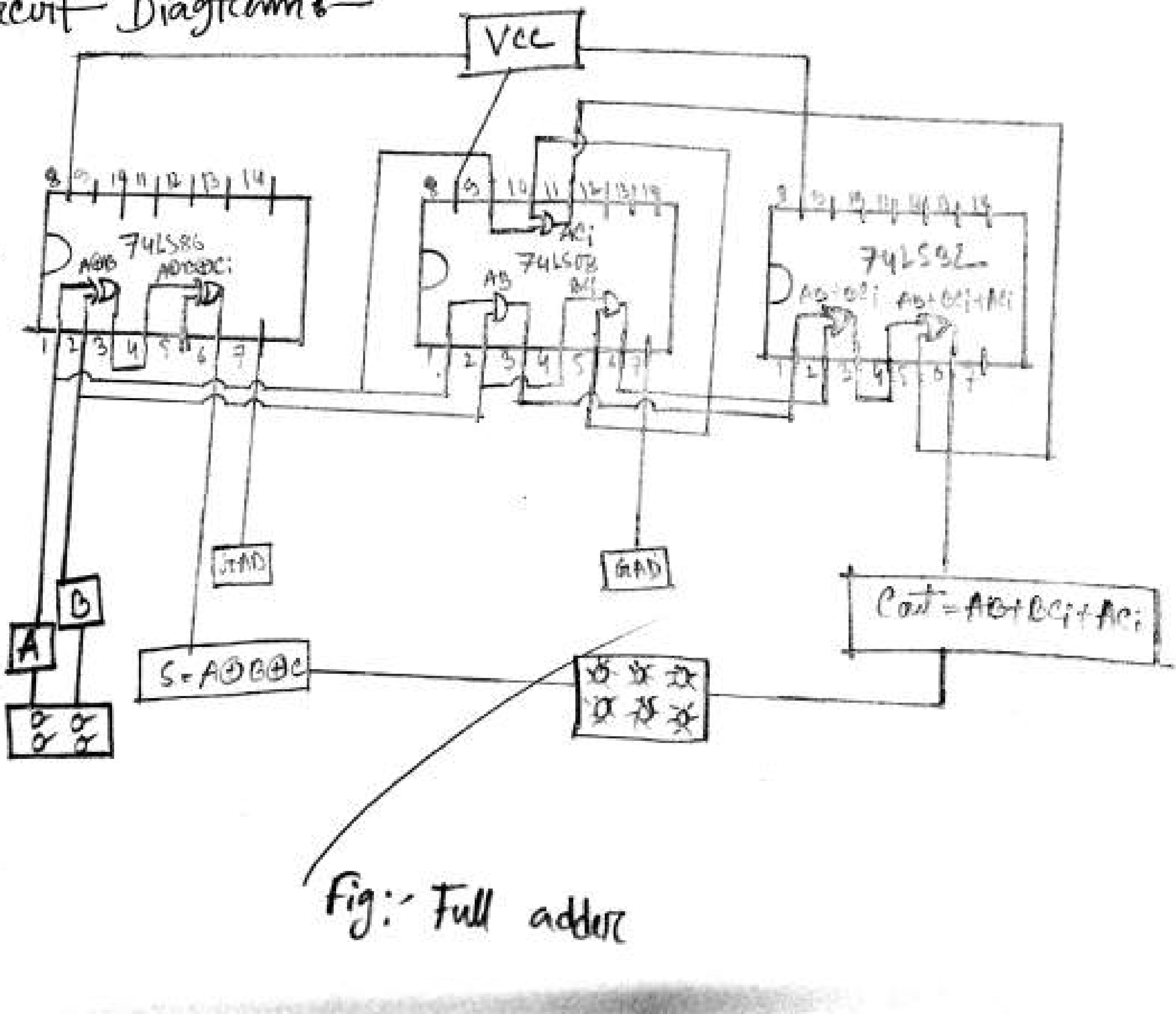


Fig:- Full adder

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Result:-

A	B	C_{in}	Sum	LED	Cout	LED
0	0	0	0	OFF	0	OFF
0	0	1	1	ON	0	OFF
0	1	0	1	ON	0	OFF
0	1	1	0	OFF	1	ON
1	0	0	1	ON	0	OFF
1	0	1	0	OFF	1	ON
1	1	0	0	OFF	1	ON
1	1	1	1	ON	1	ON

Discussion:-

From this experiment we can get the verification of the operation of Boolean expression $S = A \oplus B \oplus C_{in}$ and carry $AB + BC_{in} + AC_{in}$ using ExOR, AND, OR IC's

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Daffodil Institute of IT

Lab Report
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DSD-510209
(Digital System Design Lab)

Experiment No.	06
Experiment Name	Verify the operation of half subtractor
Date of Performance	
Date of Submission	

Submitted By

Name	**Abdur Rahim**
Student ID	**180121**
Registration No	
Section	**B**
Batch	**18th**

Submitted To

Tahmina Aktar Trisha
Faculty
Department of Computer Science & Engineering
Daffodil Institute of IT(DIIT)
64/3 & 64/4 Lake Circus (2nd-5th floor)
Mirpur Road Dhaka-1205, Bangladesh

Signature

Experiment no:- 06

Experiment name: Verify the operation of half subtractor.

Objective: The objectives of this experimental is to verify the half substractor to implement of the AND gate, NOT gate, Ex-OR gate, using specific IC's

Theory of half subtractor: In a half substrator circuit the two variables x and y represent minued and substractor respect, The two output functions are difference and borrow are symbolize as D and B respectively.

Truth table:-

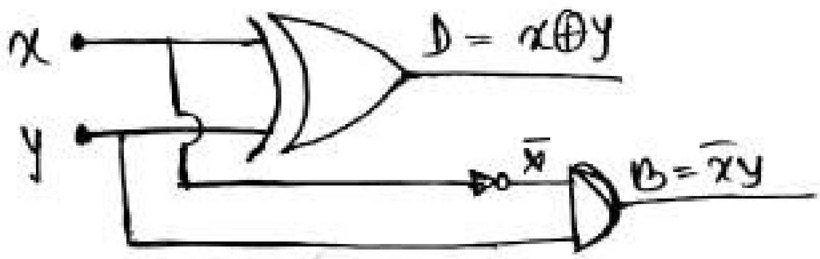
x	y	D	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Boolean expression:

$$D = \bar{x}y + x\bar{y} = x \oplus y$$

$$B = \bar{x}y$$

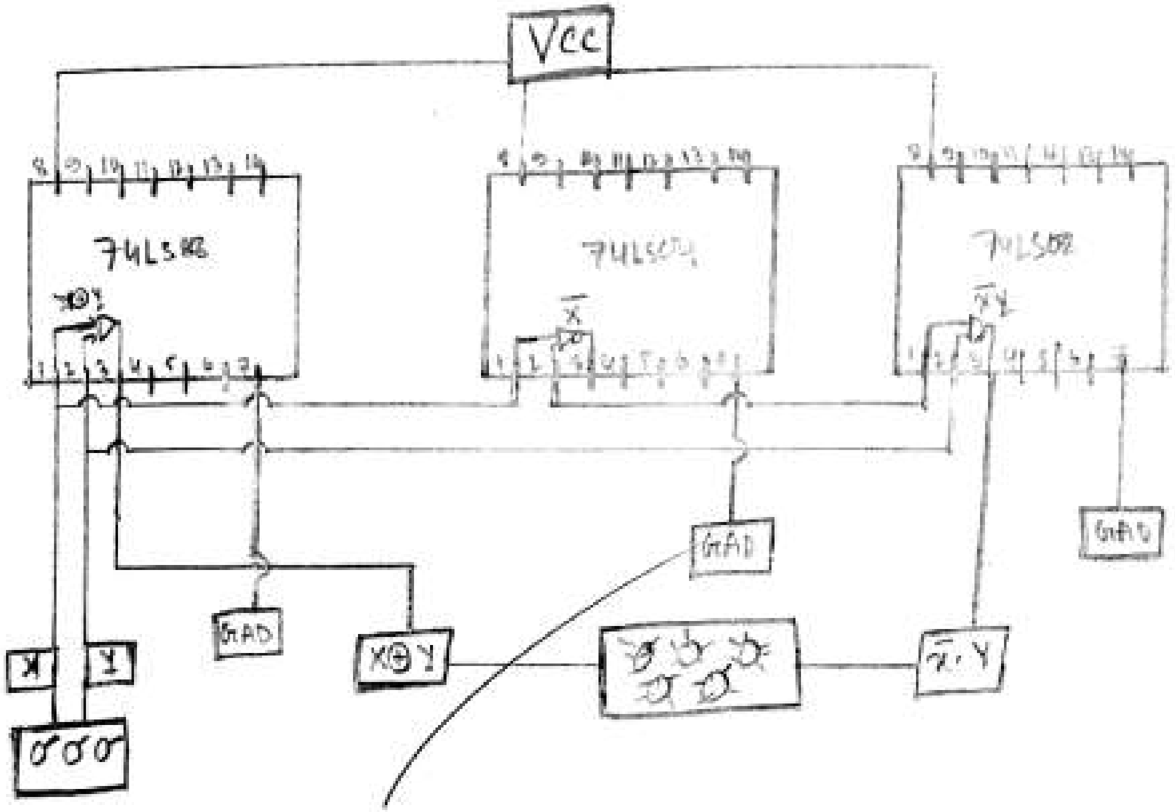
Logical circuit:



Apparatus:-

i) Ic (74LS86) Ex-OR, (74LS04) NOT, (74LS08) AND
ii) Bread Board
iii) LED
iv) connecting wire
v) Cable
vi) voltage source (+4 ... +12)

Circuit diagram :—



Result :—

X	Y	D	LED	B	LED
0	0	0	OFF	0	OFF
0	1	1	ON	1	ON
1	0	1	ON	0	OFF
1	1	0	OFF	0	OFF

Discussion :— From this experiment we can verify the operation of Boolean expression $D = X \oplus Y$ and $B = \bar{X}Y$ using (74LS86) EX-OR, (74LS04) NOT, (74LS08) AND IC's.

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Daffodil Institute of IT

Lab Report
On
DSD-510209
(Digital System Design Lab)

Experiment No.	07
Experiment Name	Verify the operation of full subtractor
Date of Performance	
Date of Submission	

Submitted By

Name	**Abdur Rahim**
Student ID	**180121**
Registration No	
Section	**B**
Batch	**18th**

Submitted To

Tahmina Aktar Trisha
Faculty
Department of Computer Science & Engineering
Daffodil Institute of IT(DIIT)
64/3 & 64/4 Lake Circus (2nd-5th floor)
Mirpur Road Dhaka-1205, Bangladesh

Signature

* Experiment No:- 07

* Experiment Name:- Verify the operation of full subtractor.

* Objective: The objective of this experiment is to verify full subtractor to implement of the Ex-OR, NOT, OR, AND gate using specific IC's.

* Operation of full subtractor: The full subtractor circuit has three inputs here two single bit data inputs s and Y. Full subtractor perform the operation of subtractor on three binary bits producing outputs for the different being and borrow-out (B-out) just like as binary adder circuits.

Truth table :

X	Y	Bin	D	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Boolean expression :

$$D = \bar{x}\bar{y}B_{in} + \bar{x}y\bar{B}_{in} + \bar{x}\bar{y}\bar{B}_{in} + xyB_{i}$$

$$= \bar{x}\bar{y}B_{in} + xyB_{in} + \bar{x}y\bar{B}_{in} + x\bar{B}_{in}\bar{y}$$

$$= B_{in}(\bar{x}\bar{y} + xy) + \bar{B}_{in}(\bar{x}y + x\bar{y})$$

$$= B_{in} \oplus x \oplus y$$

$$= x \oplus y \oplus B_{in}$$

$$B_{out} = \bar{x}\bar{y}B_{in} + \bar{x}y\bar{B}_{in} + \bar{x}yB_{in} + xyB_{in}$$

$$= \bar{x}\bar{y}B_{in} + \bar{x}yB_{in} + \bar{x}y\bar{B}_{in} + \bar{x}yB_{in} + \bar{x}yB_{in} + xyB_{in}$$

$$= \bar{x}B_{in}(y + \bar{y}) + \bar{x}y(B_{in} + B_{in}) + yB_{in}(x + \bar{x})$$

$$= \bar{x}B_{in} + \bar{x}y + yB_{in}$$

Apparatus : i) Ic (74LS86) Ex-OR
(74LS32) OR
(74LS08) AND
(74LS04) NOT

ii) LED

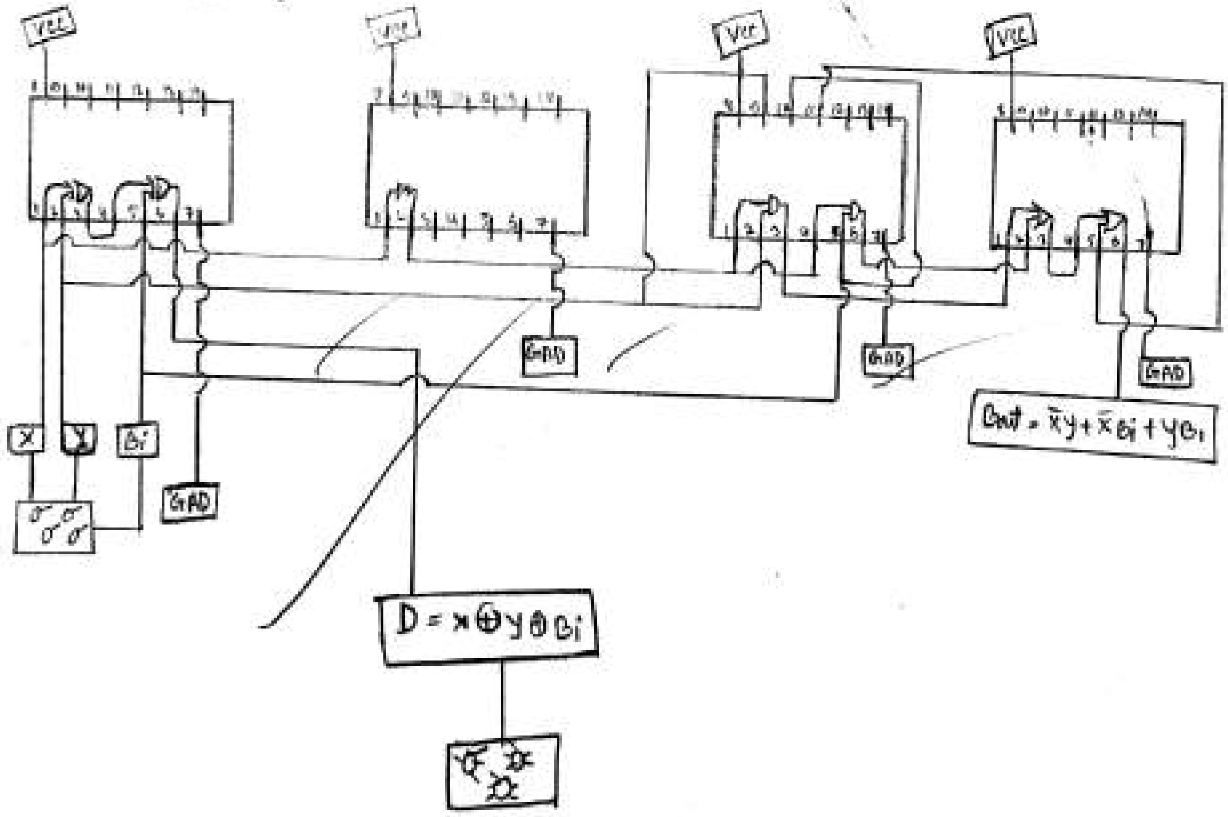
iii) Bread Board

iv) connecting wire

v) cable

vi) voltage source

Circuit specification :-



Result:

x	Y	Bin	D	LED	Bout	LED
0	0	0	0	OFF	0	OFF
0	0	1	1	ON	1	ON
0	1	0	1	ON	1	ON
0	1	1	0	OFF	1	ON
1	0	0	1	ON	0	OFF
1	0	1	0	OFF	0	OFF
1	1	0	0	OFF	0	OFF
1	1	1	1	ON	1	ON

conclusion:

A full subtractor is combinational circuit Which is used to perform subtraction of 3 bit inputs and 2 outputs. It's made of X-OR gate, NOT gate and AND gate, OR gate

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Daffodil Institute of IT

Lab Report
On
DSD-510209
(Digital System Design Lab)

Experiment No.	08
Experiment Name	Verify the operation of J-K flip flop circuit
Date of Performance	
Date of Submission	

Submitted By

Name	**Abdur Rahim**
Student ID	**180121**
Registration No	
Section	**B**
Batch	**18th**

Submitted To

Tahmina Aktar Trisha
Faculty
Department of Computer Science & Engineering
Daffodil Institute of IT(DIIT)
64/3 & 64/4 Lake Circus (2nd-5th floor)
Mirpur Road Dhaka-1205, Bangladesh

Signature

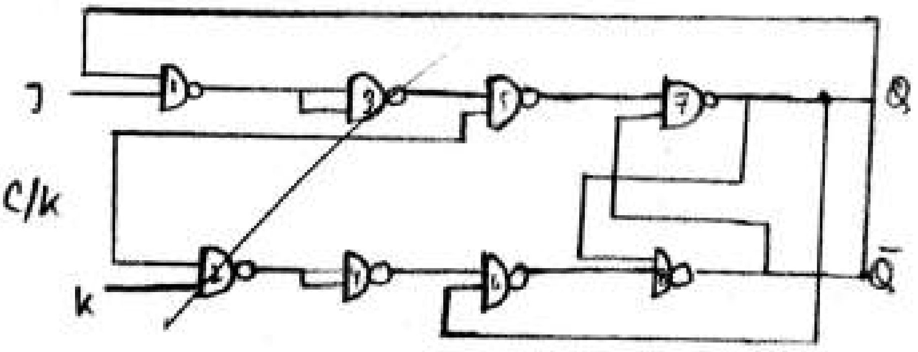
Expremont NO: 8

Exprament name: Verify the operation of J-k flip flop circuit.

Objective: The objective of this expriment is to verify the operation of J-k flip flop circuit using specfie IC.

Operation of J-k flip flop circuit: A clocked J-k flip flop that is triggred by positive going edge of the clock signal. The J-k control the state of flip flop in the same as the S and R inputs

logical symbol



Truth table:

J	K	C/K	Q $\bar{Q}$
0	0	↑	Memory
0	1	↑	1 0
1	0	↑	0 1
1	1	↑	Toggle

Apparatus:-
i) voltage source
ii) IC (74LS00 - NAND)
iii) Trainer Board
iv) connecting wire
v) switch
vi) cable
vii) Bread Board
viii) led

Circuit specification:

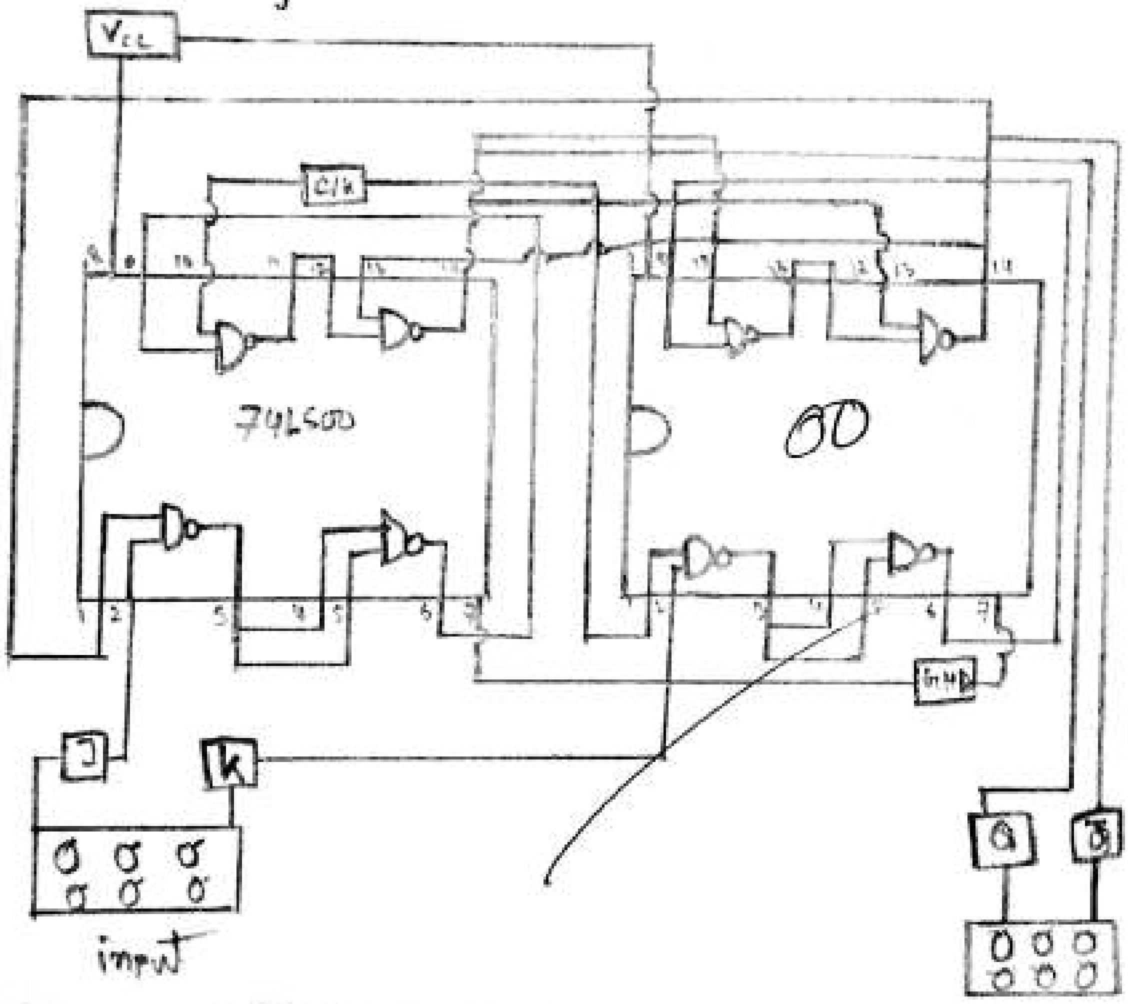


Fig: J-K flip flop circuit

Result :- In a J-k flip flop inputs are J, k, and the outputs are Q and $\bar{Q}$. In J-k flip flop we use the PGT clock pulse. If the inputs are low the LED will be memory state. If J is low and k is high the Q led will be on and $\bar{Q}$ will off. If both inputs are high the led will be toggled

Discussion: From the expriment we can know how J-k flip flop works. Also the construction of this using logic gates. We know about the clock pulse. We can use J-k flip flop in practical life for constructing circuit.

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Lab Report
On
DSD-510209
(Digital System Design Lab)

Experiment No.	09
Experiment Name	Verify the operation of Decoder circuit
Date of Performance	
Date of Submission	

Submitted By

Name	**Abdur Rahim**
Student ID	**180121**
Registration No	
Section	**B**
Batch	**18th**

Submitted To

Tahmina Aktar Trisha
Faculty
Department of Computer Science & Engineering
Daffodil Institute of IT(DIIT)
64/3 & 64/4 Lake Circus (2nd-5th floor)
Mirpur Road Dhaka-1205, Bangladesh

Signature

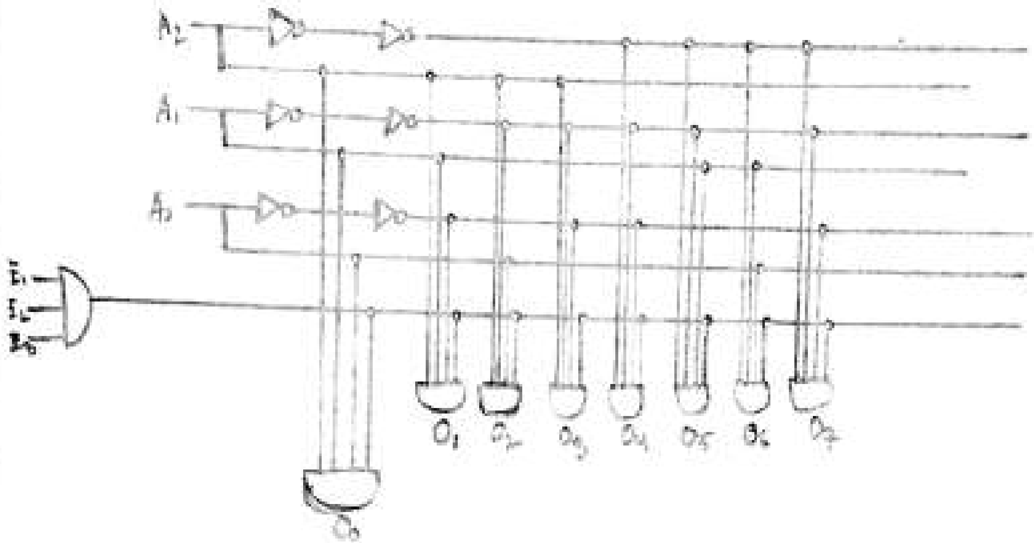
Experiment No:- 09

Experiment Name: Verify the operation of Decoder circuit.

Objective: The objective of the experiment is to verify the operation of Decoder circuit using specific IC's

Operation of Decoder circuit: A decoder is a logic circuit that accepts a set of inputs that represent a binary number and activates only the output that corresponding to that input number. In other words, a decoder circuit looks like at it's input determines which binary number is present there.

Logical Diagram :-



Logical diagram of 74LS138 Decoder

Boolean expression —

If the inputs of Decoder is A, B and C then the output will be $O_0, O_1, O_2, O_3, O_4, O_5, O_6, O_7$. Then the Boolean expression of the decoder circuit will be

$O_0 = \bar{A}\bar{B}\bar{C}$

$O_1 = \bar{A}B\bar{C}$

$O_2 = \bar{A}BC$

$O_3 = AB\bar{C}$

$O_4 = \bar{A}\bar{B}C$

$O_5 = A\bar{B}C$

$O_6 = \bar{A}BC$

$O_7 = ABC$

Truth table:-

Input			Output							
C	B	A	O_7	O_6	O_5	O_4	O_3	O_2	O_1	O_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

Instruments:-

i) Ic - 74LS138
ii) voltage sorce (+5v to +12v)
iii) Bread Board
iv) Trainer Board
v) switch
vi) cable
vii) LED
viii) connecting wire

Circuit specification :-

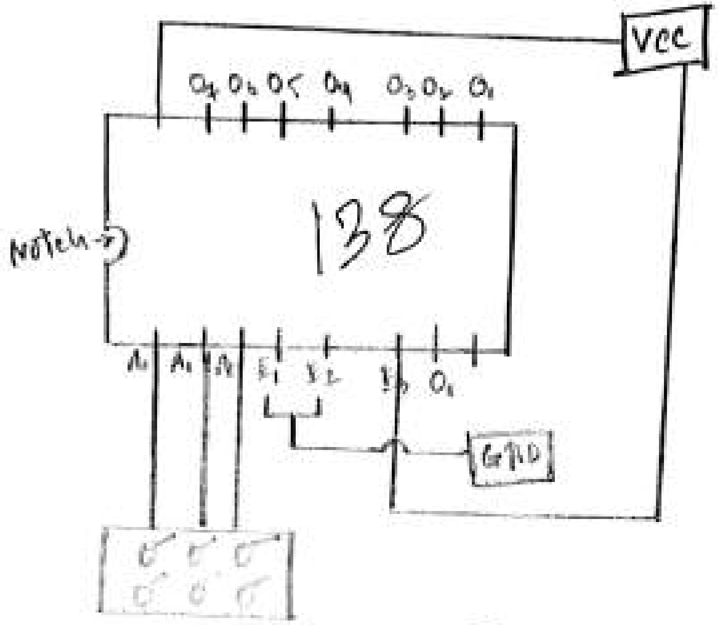


Fig:- Decoder circuit

Result :-

The LED will be on when output is 1. LED will be off when output is 0.

Conclusion :-

Decoder is realized in data flow model of architecture and the output waveform timing analysis is also obtained